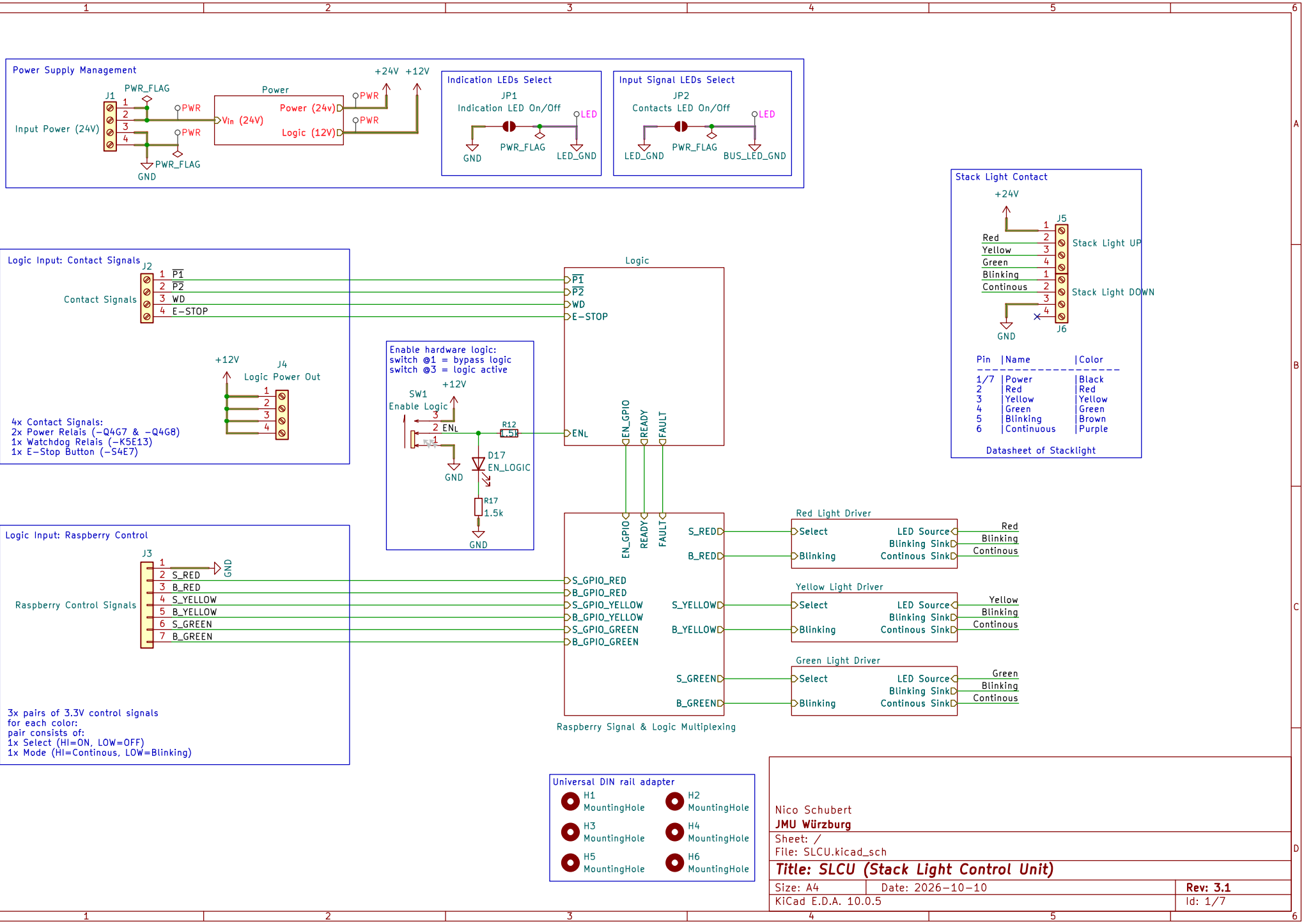
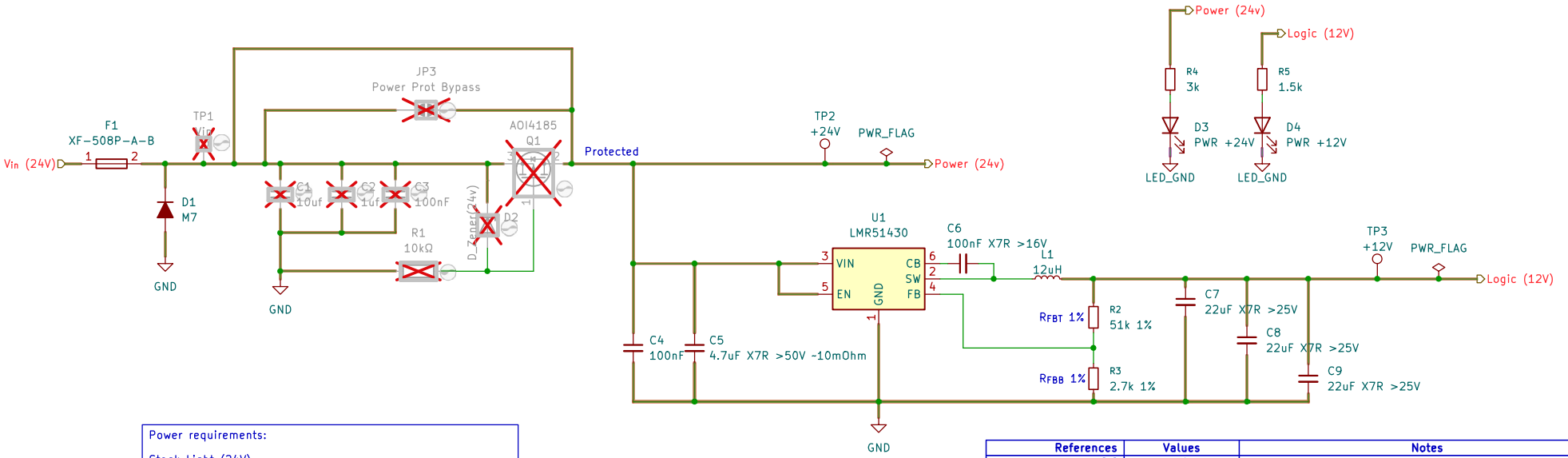


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JMU Würzburg

Sheet: /
File: SLCU.kicad_sch

Title: SLCU (Stack Light Control Unit)

Size: A4	Date: 2026-10-10	**Rev: 3.1**
KiCad E.D.A. 10.0.5		Id: 1/7



Power requirements:

Stack Light (24V):
I = 0.06A à 24V => P = 1.44W

LEDs (12V):
4x Relais
6x GPIO
7x Logic Status
2x Power
---
19 LEDs à ~10mA => I < 200mA (12V) => P = 2.4W
à ~15mA => I < 300mA (12V) => P = 3.6W

Logic ICs (neglectable)(12V):
P_{sum} < 1mW

Mosfets (12V):
1x Inverters:
à I = 12V / 47k = 0.255mA
à P = 3.064mW
16x Gate Drive (absolute) Peak current:
9x Logic
7x GPIO
(3*2)x Stack Light drive
à I = 12V / 1k = 12mA => I_{sum} = 264mA
à P = 144mW => P_{sum} = 3.17W

TOTAL:
$I_{12V)TOTAL}$ = 0.564 A
$P_{12V)TOTAL}$ = 6.771 W

References	Values	Notes
V_{in} [V]	**24**	
V_{out} [V]	**12**	
I_{out} [A]	**0.75**	
f_{sw} [kHz]	**500**	500 kHz or 1100 kHz
K_{ind} [%]	**60**	
R_{FBT} [kOhm]	**51**	10k < R_{FBT} < 100k << 1M
L [uH]	**12**	
$\Delta V_{out,ESR}=\Delta V_{out,C}$ [V]	**0.1**	
Calculations:		
Duty Cycle [%]	*50.0*	
V_{ref} [V]	*0.6*	
R_{FBT} / R_{FBB}	*19.000*	(V_{out} − V_{ref}) / V_{ref}
R_{FBB} [kOhm]	*2.684*	R_{FBT} * V_{ref} / (V_{out} − V_{ref})
E24 R_{FBB} [kOhm]	*2.7*	$V_{out(true)}$ = V_{ref} * (1 + R_{FBT} / R_{FBB})
actual V_{out} [V]	*11.933*	Plug in above to perform calculations with true V_{out}
I_{FB} [mA]	*0.224*	V_{out} / (R_{FBT} + R_{FBB})
P_{FB} [mW]	*2.682*	V_{out}^2 / (R_{FBT} + R_{FBB})
L_{min} [uH]	*26.7*	V_{out} * (V_{in} − V_{out}) / (V_{in}*f_{sw} * I_{out}*K_{ind})
ΔI_L [A]	*1.000*	V_{out} * (V_{in} − V_{out}) / (V_{in}*f_{sw} * L)
$I_{IC,max}$ [A]	*4.260*	$I_{LIM,H}$ − ΔI_L / 2 > I_{out}
$I_{sw,max}$ [A]	*1.250*	I_{out} + ΔI_L / 2
ESR_{Cout} [Ohm]	*0.100*	$\Delta V_{out,ESR}$ / ΔI_L
C_{out} [uF]	*2.500*	ΔI_L / (8 * f_{sw} * $\Delta V_{out,C}$)

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Sheet: /Power/
File: power.kicad_sch

Title: SLCU (Stack Light Control Unit)

Size: A4	Date: 2026-10-10	**Rev: 3.1**
KiCad E.D.A. 10.0.5		Id: 2/7

Singal Describtion:

- $\overline{P1}$ and $\overline{P2}$: Power relais (-Q4G7 & -Q4G8)
  - => HIGH = Contacts opend and bus unpowered
  - => HiZ = Contacts closed and bus powered
- WD: Watchdog relais (-K5E13)
  - => HIGH = Watchdog tripped and Load bus unpowered
  - => HiZ = All OK and Load bus powered
- E-STOP: (Emergency-) Stop button
  - => HIGH = Stop button pressed
  - => HiZ = Stop button not pressed
- $\overline{EN_L}$: $\overline{Enable}$/Bypass static logic control
  - => HIGH = directly forward Rapsberry commands bypassing Logic
  - => LOW = enable Hardware Logic

Help Signals:
F := in Fault configuration
READY := System Ready to be started

Logic Rules:
// all contacts need to be in their normal/unpowered state
// WD is normally HIGH when unpowered
READY = E-STOP ∧ P1 ∧ P2 ∧ WD

All individual Fault conditions:
F1 = (E-STOP ∧ ¬(P1 ∧ P2 ∧ WD)) // E-STOP but Relais not opening
F2 = ¬(P1 ↔ P2) // asymmetry in P1 and P2
F3 = (WD ∧ ¬(P1 ∨ P2)) // WD triggered when bus powered
F4 = (WD ∧ (P1 ∧ P2)) // WD stuck closed

F = F1 ∨ F2 ∨ F3 ∨ F4

// Results in the following truth Table for F

E-STOP	WD	P2	P1	F	F1	F2	F3	F4	Chip/Output	
0	0	0	0	0	0	0	0	0	U2/0	// NOMINAL ON
0	0	0	1	0	0	1	0	0	U2/1	
0	0	1	0	1	0	1	0	0	U2/2	
0	0	1	1	1	0	0	0	1	U2/3	
0	1	0	0	1	0	0	1	0	U2/4	
0	1	0	1	1	0	1	0	0	U2/5	
0	1	1	0	1	0	1	0	0	U2/6	
0	1	1	1	0	0	0	0	0	U2/7	// READY
1	0	0	0	1	1	0	0	0	U3/0	
1	0	0	1	1	1	1	0	0	U3/1	
1	0	1	0	1	1	1	0	0	U3/2	
1	0	1	1	1	1	1	0	0	U3/3	
1	1	0	0	1	1	1	0	1	U3/4	
1	1	0	1	1	1	1	0	0	U3/5	
1	1	1	0	1	1	1	0	0	U3/6	
1	1	1	1	0	0	0	0	0	U3/7	// STOPPED

// Conjunctive normal form therefor simplifies to
F = (P1 ∧ P2 ∧ WD) ∨ (¬P1 ∧ ¬P2 ∧ WD ∧ E-STOP)

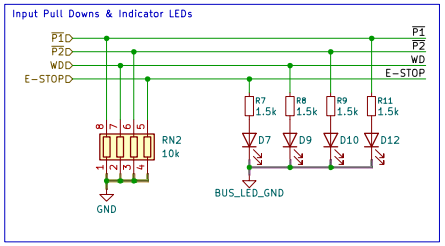
=> F = (¬P1 ∨ ¬P2 ∨ WD) ∧ (P1 ∨ P2 ∨ WD ∨ E-STOP)

// When Using only a 3-to-8 Decoder using signals [WD, P2, P1]
// then the following logic aplies:

NOMINAL ON = out000 ∧ E-STOP
READY = out111 ∧ E-STOP
STOPPED = out111 ∧ E-STOP
F1 = out111 ∧ E-STOP

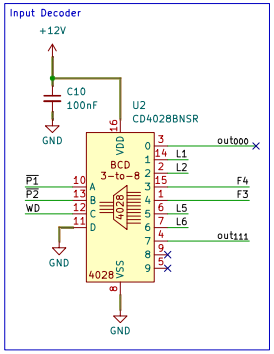
// Using De Morgans whe can avoid AND-Gates by "double" Inverting

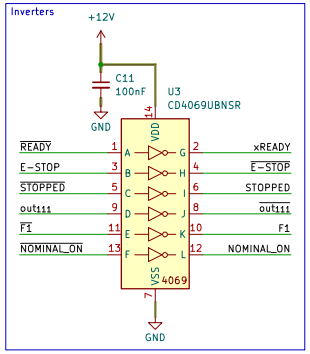
out000 = P1 ∨ P2 ∨ WD
NOMINAL ON = out000 ∨ E-STOP = P1 ∨ P2 ∨ WD ∨ E-STOP
READY = out111 ∨ E-STOP
STOPPED = out111 ∨ E-STOP
F1 = out111 ∨ E-STOP

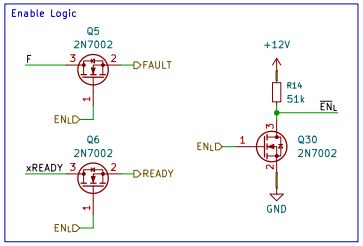


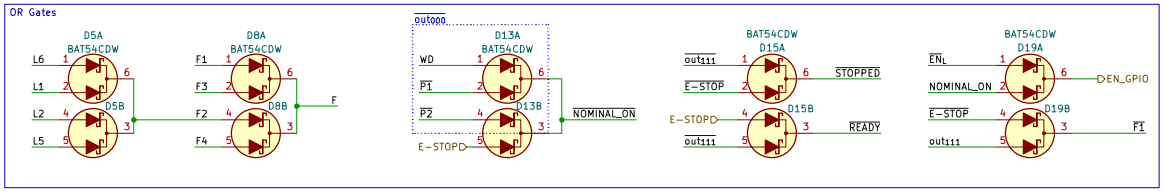
Note:
All the CD4xxx family ICs register a high-level as low as -3V below 12V.
Therefor OR-ing together signals via diodes can be achieved as long as the accumulated voltage drop along a signal is less than the 3V minus a safty margin.
With Schottky diods of V_F of about 0.7V a line can handle up to:

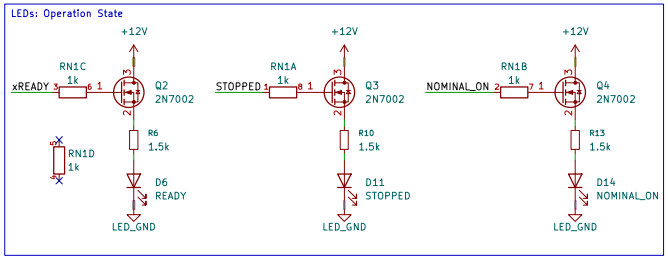
(3V - margin) / V_F = (3V - 0.5V) / V_F = 3.571 >= 3 diodes in series

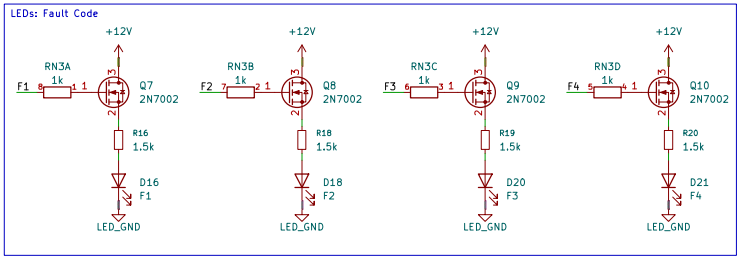












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Sheet: /Logic/
File: logic.kicad_sch

Title: SLCU (Stack Light Control Unit)

Size: A3	Date: 2026-10-10	**Rev: 3.1**
KiCad E.D.A. 10.0.5		Id: 3/7

$I_{max,Burst}$ = 7*(12V − 2V) / 1500Ω + 6*12V / 1000Ω = 118.667mA
$I_{max,Cont}$ = 7*(12V − 2V) / 1500Ω + 6*12V / 11000Ω = 53.212mA

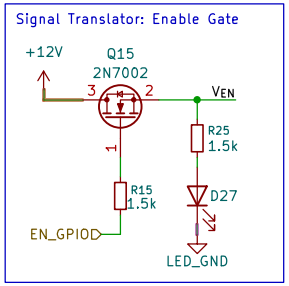
max Burst PD = $R_{DS(on)}$ * $I_{max,Burst}^2$ = 28.164mW @ $R_{DS(on)}$=2Ω
max Cont. PD = $R_{DS(on)}$ * $I_{max,Cont}^2$ = 5.663mW @ $R_{DS(on)}$=2Ω

Logic Rules for Stack Light Illumination

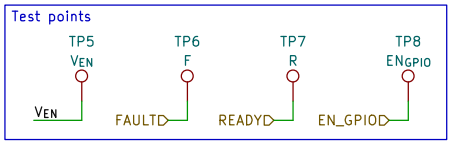
	NORMALY_ON "N"	FAULT "F"	READY "R"	
S_R	π_{RS}	1	0	$S_R = N \wedge \pi_{RS} \vee F$
B_R	π_{RB}	1	–	
S_Y	π_{YS}	0	0	$S_Y = N \wedge \pi_{YS}$
B_Y	π_{YB}	–	–	
S_G	π_{GS}	0	1	$S_G = N \wedge \pi_{GS} \vee R$
B_G	π_{GB}	–	1	

=> $B_X = N \wedge \pi_{XB}$, $X \in \{R, Y, G\}$

Signal Translator: Enable Gate

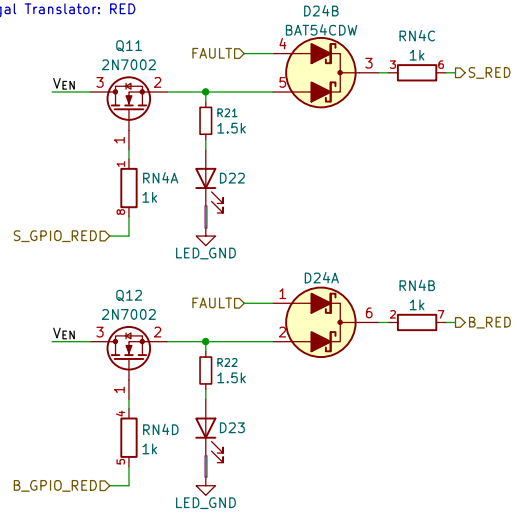


Test points

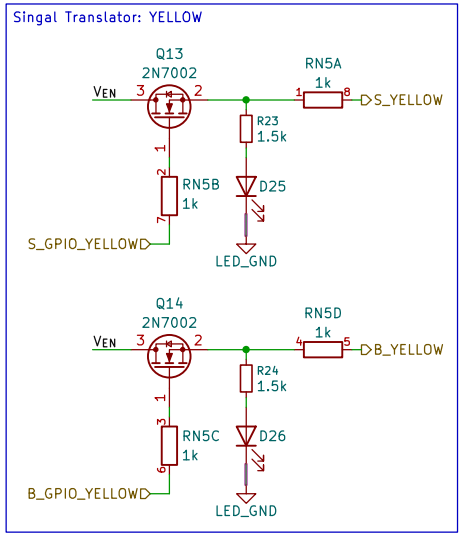


$I_{max,Color}$ = (12V − 2V) / 1500Ω + 12V / 1000Ω = 18.667mA
max PD = $R_{DS(on)}$ * $I_{max,Color}^2$ = 0.697mW @ $R_{DS(on)}$=2Ω
max $R_{DS(on)}$ = 200mW / (18.667mA)² = 574Ω

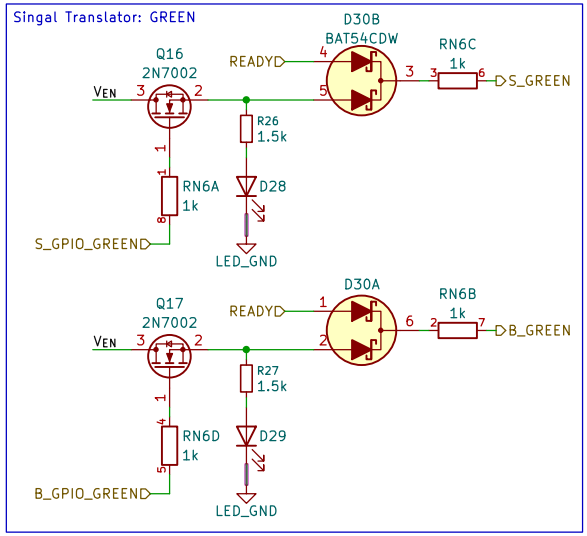
Singal Translator: RED



Singal Translator: YELLOW



Singal Translator: GREEN



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Sheet: /Raspberry Signal & Logic Multiplexing/
File: GPIO_MUX.kicad_sch

Title: SLCU (Stack Light Control Unit)

Size: A4	Date: 2026-10-10	**Rev: 3.1**
KiCad E.D.A. 10.0.5		Id: 4/7

max PD = $R_{DS(on)}$ * (0.043A)² = 4mW @ $R_{DS(on)}$=2Ω
max $R_{DS(on)}$ = 200mW / (50mA)² = 80Ω

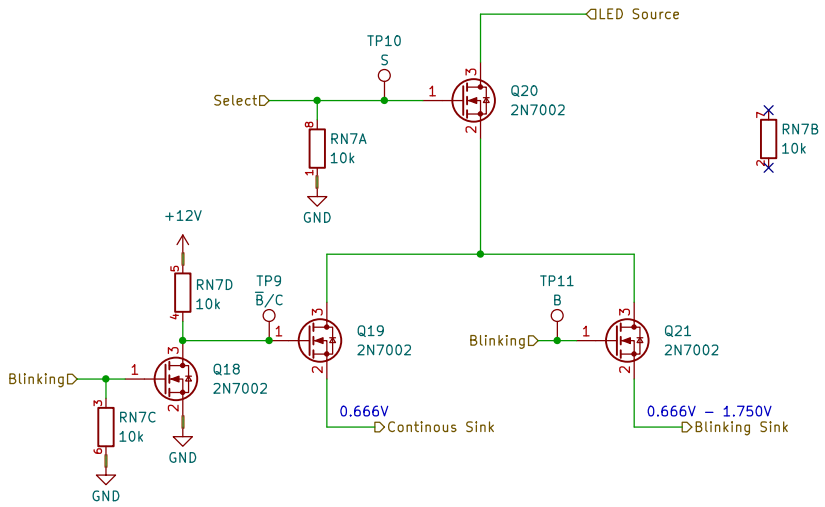
Both Sink connections/wires (Continous and Blinking) are internally protected against reverse currentflow and therefor dont impose a risk of short circuits when connected together.

The Continously on Sink overrules the Blinking Sink in the case that both are connected to an LED at the same time.

Stack light voltages:

Continous	C	0.666 V
Blinking	B	0.666 V
		(1.75 V +−0.10 V)?
Red line	R	9.55 V +−0.05 V
Yellow line	Y	8.80 V +−0.05 V
Green line	G	7.52 V +−0.03 V

Stack light currents:
I_{CR} = 40 mA +−3 mA
I_{CY} = 31 mA +−4 mA
I_{CG} = 12 mA +−3 mA



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Sheet: /Red Light Driver/
File: LED_controller.kicad_sch

Title: SLCU (Stack Light Control Unit)

Size: A4	Date: 2026-10-10	**Rev: 3.1**
KiCad E.D.A. 10.0.5		Id: 5/7

max PD = $R_{DS(on)}$ * (0.043A)² = 4mW @ $R_{DS(on)}$=2Ω
max $R_{DS(on)}$ = 200mW / (50mA)² = 80Ω

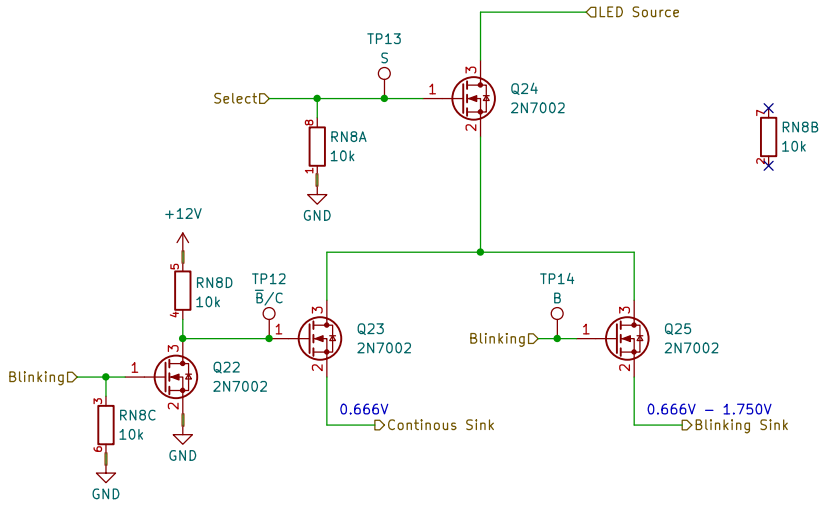
Both Sink connections/wires (Continous and Blinking) are internally protected against reverse currentflow and therefor dont impose a risk of short circuits when connected together.

The Continously on Sink overrules the Blinking Sink in the case that both are connected to an LED at the same time.

Stack light voltages:

Continous	C	0.666 V
Blinking	B	0.666 V
		(1.75 V +−0.10 V)?
Red line	R	9.55 V +−0.05 V
Yellow line	Y	8.80 V +−0.05 V
Green line	G	7.52 V +−0.03 V

Stack light currents:
I_{CR} = 40 mA +−3 mA
I_{CY} = 31 mA +−4 mA
I_{CG} = 12 mA +−3 mA



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Sheet: /Yellow Light Driver/
File: LED_controller.kicad_sch

Title: SLCU (Stack Light Control Unit)

Size: A4	Date: 2026-10-10	**Rev: 3.1**
KiCad E.D.A. 10.0.5		Id: 6/7

max PD = $R_{DS(on)}$ * (0.043A)² = 4mW @ $R_{DS(on)}$=2Ω
max $R_{DS(on)}$ = 200mW / (50mA)² = 80Ω

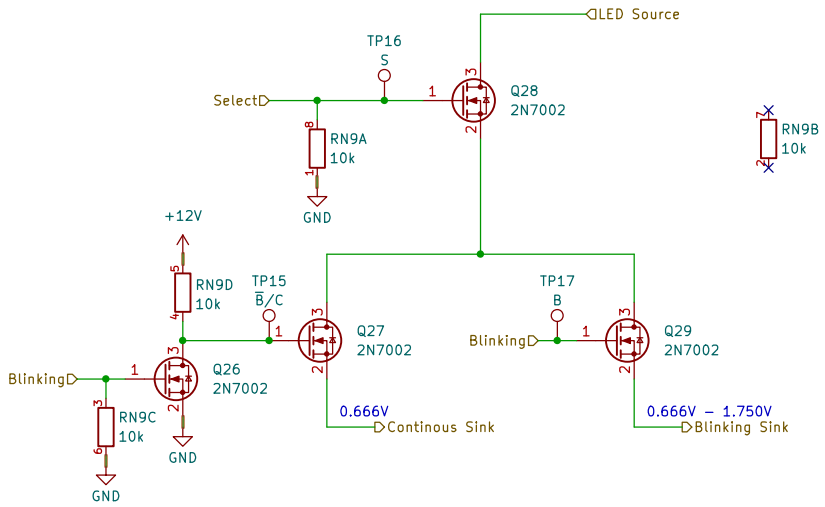
Both Sink connections/wires (Continous and Blinking) are internally protected against reverse currentflow and therefor dont impose a risk of short circuits when connected together.

The Continously on Sink overrules the Blinking Sink in the case that both are connected to an LED at the same time.

Stack light voltages:

Continous	C	0.666 V
Blinking	B	0.666 V
		(1.75 V +−0.10 V)?
Red line	R	9.55 V +−0.05 V
Yellow line	Y	8.80 V +−0.05 V
Green line	G	7.52 V +−0.03 V

Stack light currents:
I_{CR} = 40 mA +−3 mA
I_{CY} = 31 mA +−4 mA
I_{CG} = 12 mA +−3 mA



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Sheet: /Green Light Driver/
File: LED_controller.kicad_sch

Title: SLCU (Stack Light Control Unit)

Size: A4	Date: 2026-10-10	**Rev: 3.1**
KiCad E.D.A. 10.0.5		Id: 7/7